

Problem-Solution Fit

Date	25 June 2026
Team ID	
Project Name	Human Development Index (HDI) Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience .

1. CUSTOMER SEGMENT(S) [CS]

- Government policymakers
- Researchers and analysts
- Students studying Human Development
- Development organizations

2. PROBLEMS / PAINS [PR]

+ ITS FREQUENCY

- Manual HDI calculation is time-consuming.
- Analyzing multiple development indicators is difficult.
- Lack of a simple tool for quick HDI prediction.
- Frequently faced while comparing development across countries.

3. TRIGGERS TO ACT [TR]

- Need to evaluate a country's development.
- Government planning and policy making.
- Academic research and comparative analysis.
- Requirement for quick HDI prediction using available indicators.

4. EMOTIONS BEFORE / AFTER [EM]

Before:

- Confused
- Concerned about accuracy
- Time-consuming analysis

After:

- Confident
- Better decision-making
- Faster and easier HDI prediction

5. AVAILABLE SOLUTIONS [AS]

Pros

- UNDP publishes official HDI reports.
- Government reports provide development statistics.
- Online datasets are publicly available.

Cons

- HDI must be calculated manually.
- Reports are not interactive.
- No instant prediction from user inputs

6. CUSTOMER LIMITATIONS [CL]

Example: Budget, Devices

- Limited knowledge of HDI calculation.
- Lack of technical expertise in Machine Learning.
- Manual analysis requires significant time.
- Dependence on official reports and datasets.

7. BEHAVIOR + ITS INTENSITY [BE]

- Users frequently analyze development indicators.
- Researchers compare HDI across countries.
- Policymakers use development data for planning.
- Users require quick and reliable predictions.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Flask Web Application
- Government Development Portals
- Educational Websites
- Research Platforms

OFFLINE

- Government Reports
- Universities
- Research Institutions
- Development Organizations

9. PROBLEM ROOT / CAUSE [RC]

- HDI depends on multiple socio-economic indicators.
- Manual calculation is complex.
- No simple prediction system for non-technical users.
- Large amounts of development data require analysis.

10. YOUR SOLUTION [SL]

A **Flask-based Machine Learning application** that predicts the **Human Development Index (HDI)** using **Life Expectancy, Mean Years of Schooling, GNI per Capita, Internet Users, and Country**. The system instantly predicts the **HDI score** and classifies it into **Low, Medium, High, or Very High Human Development**, helping users analyze development efficiently.